BioDYM
User Manual

Material Flow Analysis with Dynamic Stock
and Decay Modeling

Version 1.2.1

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Built on the ODYM Framework

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Contents

What's New in v1.2

This manual was originally written for BioDYM v1.0 (February 2026). The chapters that follow have been partially updated for v1.2.1 (June 2026), but several new features are not yet fully documented. This chapter gives a concise overview of what has changed so you can get started without waiting for a complete manual revision.

bioDYM SystemDefiner (new in v1.2)

The biggest addition is the **bioDYM SystemDefiner** — an interactive web application for defining case studies visually without editing files directly.

```
uv run python -m systemdefiner
```

Opens at `http://localhost:8001`. You can define processes, flows, transfer coefficients, DSM/FOMP/LFG parameters, scenarios, and Monte Carlo settings through a browser interface, then export a `config.yaml` file that the engine reads directly.

Tutorial case studies (T01–T07) are included in the repository under `01_data/01_input/case_studies/` and open automatically when you start the SystemDefiner.

YAML-only workflow (new in v1.2)

The engine no longer requires an Excel file. If you export a `config.yaml` from the SystemDefiner (or write one manually), you can pass it directly as the input:

```
input_file = "01_data/01_input/case_studies/MyStudy/config.yaml"
```

The notebook and Voìlà dashboard both accept `.yaml` paths in place of `.xlsm`.

Voìlà dashboard (new in v1.2)

A self-contained interactive dashboard is available as an alternative to the Jupyter notebook:

```
uv run voila 01_BioDYM_Dashboard.ipynb
```

The dashboard provides the same file-path input, results, and visualizations in a clean interface without exposing the underlying code.

New process logic types (new in v1.2)

Three new process logic types are available (see Chapter 5 for existing types):

Process Logic	Behavior
LFG	Landfill gas model — N-pool first-order decay (UNFCCC AM-Tool-04); outputs CH_4 and CO_2-C. See Chapter 12.
FlowCap	Capacity-limited routing — caps an outflow at a time-series maximum and routes excess elsewhere.
BOM Assembler	Bill-of-materials composition tracking — assembles a product from defined input fractions.

Table 1: New process logic types added in v1.2.

Other changes since v1.0

- **DSM multi-category**: Each DSM process now supports multiple product cohorts with independent lifetime distributions (Normal, LogNormal, Weibull, Fixed) and per-category inflow splits.
- **Sankey export**: Results can be exported to HTML, JSON, eSankey, and SankeyMATIC formats from notebook Section 5.2.
- **Initial stock engine**: Improved cohort-based age distribution with depletion modelling.
- **Element naming**: Carbon is now called `TC` (Total Carbon from CHNS analysis) throughout. Legacy files using `CC` continue to work via an automatic fallback.
- **Getting Started guide**: `GETTING_STARTED_FROM_ZERO.md` at the repository root is the recommended starting point for new users.

Note

The chapters that follow document the v1.0 feature set, with targeted updates where content has changed. A fully revised manual covering the SystemDefiner, YAML workflow, and new process types in depth is planned for v1.3.

Chapter 1

Introduction

1.1 What is BioDYM?

BioDYM is a Material Flow Analysis (MFA) framework built on top of the Open Dynamic Material Systems Model (ODYM). It provides a complete workflow for modeling material flows, stocks, and transformations in bio-based and industrial systems.

BioDYM extends ODYM with:

- **Dynamic Stock Modeling (DSM)** — Cohort-based stock tracking with configurable lifetime distributions (Normal, Weibull, LogNormal, etc.)
- **First-Order Multi-Pool decay (FOMP)** — Two-pool organic matter decomposition (labile and recalcitrant fractions)
- **Monte Carlo Simulation** — Uncertainty quantification with multiple distribution types and sensitivity analysis
- **Scenario Analysis** — Define and compare parameter modifications across multiple scenarios
- **Excel-based Configuration** — All model inputs defined in a single Excel workbook
- **Interactive Visualization** — Sankey diagrams, dynamics plots, stock analysis, and publication-quality export

1.2 The Element Concept

BioDYM tracks multiple *elements* simultaneously across all flows and stocks. Elements represent mass-conserved quantities that are tracked in parallel:

Element	**Abbreviation**	**Description**
Material	material	Total mass (always the first element)
Water Content	WC	Mass of water in the flow
Dry Matter	DM	Mass of dry matter (material = WC + DM)
Total Carbon	TC	Mass of carbon in the dry matter (legacy files may use `CC` — both are supported)

Table 1.1: Default element configuration for biomass systems.

Elements can be configured for any application. For example, a metals system might use `[material, Fe, Cu, Al]`, while a packaging system might use `[material, PET, PE, PP]`.

1.2.1 Element Hierarchy

Elements can have parent–child relationships. In the default biomass configuration:

- DM is defined as a *percentage of material*
- CC is defined as a *percentage of DM* (not of material)

This hierarchy is configured in the Excel workbook and automatically maintained by the calculation engine.

1.3 System Architecture

A BioDYM model consists of:

Processes Nodes in the system (e.g., Forestry, Processing, Use Phase)

Flows Directed connections between processes, carrying mass of each element over time

Stocks Material accumulated within a process

Transfer Coefficients (TCs) Parameters controlling how material is distributed from a process to its outflows

Parameters All model inputs: flow data, compositions, TC values, DSM/FOMP configurations

Process 0 is always the *system boundary* (environment). Flows from Process 0 into the system are inputs; flows to Process 0 are outputs.

1.4 Process Types

Each process has a *logic* that determines how it handles material:

Process Logic	Behavior
Input	System entry point (boundary process)
Splitter	Distributes material to outflows using TCs; preserves element composition
Transformer	Distributes material using element-specific TCs; changes composition
Pass-through	Passes all inflow directly to a single outflow
DSM	Dynamic stock with cohort-based lifetime tracking (multiple categories supported)
FOMP	First-order multi-pool organic matter decay (labile + recalcitrant pools)
LFG	Landfill gas N-pool first-order decay model (UNFCCC AM-Tool-04)
FlowCap	Capacity-limited flow routing with time-series cap values
BOM Assembler	Bill-of-materials product composition assembly
InitialStock	Process seeded with a pre-existing stock at model start

Table 1.2: Available process logic types in BioDYM.

Chapter 2

Installation & Setup

2.1 Requirements

- Python 3.12 or later
- `uv` package manager (https://docs.astral.sh/uv/)
- Git (for cloning the repository)
- Microsoft Excel or LibreOffice Calc (for editing input files)

2.2 Installation

1. **Clone the repository:**

```
git clone https://github.com/JScholz-tech/Biodym_JS.git
cd Biodym_JS
```

2. **Install dependencies with uv:**

```
uv sync
```

This creates a virtual environment in `.venv/` and installs all dependencies from `pyproject.toml`.

3. **Verify the installation:**

```
uv run python -m pytest 04_tests/ -v
```

All tests should pass. If any fail, check the Troubleshooting appendix (Appendix C).

2.3 Defining Your System (bioDYM SystemDefiner)

The recommended starting point is the **bioDYM SystemDefiner** — a browser-based application for configuring case studies without editing files:

```
uv run python -m systemdefiner
```

Opens at http://localhost:8001. Tutorial studies T01–T07 are pre-loaded. Define your processes, flows, and parameters, then click **Export YAML** to save a `config.yaml` for the engine.

2.4 Running the Analysis

BioDYM can be run in two ways:

2.4.1 Option A — Voìlà dashboard (recommended for users)

```
uv run voila 01_BioDYM_Dashboard.ipynb
```

A clean interactive interface opens in the browser. Enter the path to your `config.yaml` or `.xlsm` file and click Run.

2.4.2 Option B — Jupyter notebook (recommended for developers)

```
uv run jupyter lab
```

Open `00_BioDYM_Workflow.ipynb`. This exposes all intermediate results and is the right choice when you want to inspect or extend the analysis.

Tip

The notebook is paired with a Python script `00_BioDYM_Workflow.py` via Jupytext. If you edit the `.py` file, synchronize with:

```
uv run jupytext --to notebook 00_BioDYM_Workflow.py
```

2.5 Project Structure

```
BioDYM/
  00_BioDYM_Workflow.ipynb        Main notebook
  00_BioDYM_Workflow.py           Paired Python source (Jupytext)
  01_BioDYM_Dashboard.ipynb       Voila dashboard
  01_data/
    01_input/
      template/                   Blank Excel Systemmanager
      case_studies/T01_...T07_*/   Tutorial studies (YAML)
    02_output/                    Calculation results, exports
  02_src/
    engine/                       Solver, DSM, FOMP, LFG, MC engines
    plotting/                     All visualization modules
    systemdefiner/                Web app (SystemDefiner)
    system_setup.py               Model initialization
    data_loader.py                Excel/YAML data loading
  03_studies/                     Published case studies
  04_tests/                       Automated test suite
  05_docs/                        Documentation & manual
  06_framework/                   ODYM base framework
```

Chapter 3

The BioDYM Workflow

The BioDYM workflow is organized into four sequential steps, each corresponding to a section in the Jupyter notebook.

3.1 Overview

1. **Setup & Data Loading** — Load the Excel workbook, validate input data, initialize the MFA system
2. **Calculation & Validation** — Run the iterative solver, check mass balance, verify results
3. **Visualization** — Explore results with interactive plots (Sankey, dynamics, stocks, composition)
4. **Export & Advanced Analysis** — Export results, run Monte Carlo simulation, run scenarios

3.2 Step 1: Setup & Data Loading

The notebook begins by loading your Excel configuration file:

```
# Set path to your Excel input file
input_file = "01_data/01_input/your_system.xlsm"

# Load and validate
model_scope = define_model_scope(input_file)
mfa_system = initialize_mfa_system(model_scope)
mfa_system = define_flows_and_parameters(mfa_system, input_file)
```

During loading, BioDYM:

- Reads all Excel sheets and validates their structure
- Creates Process, Flow, Stock, and Parameter objects
- Interpolates time series data to fill gaps between defined years
- Normalizes transfer coefficients so outflows sum to 100% per process
- Calculates element compositions from the defined fractions

Note

If data points are only defined for certain years (e.g., 2020 and 2030), BioDYM automatically interpolates linearly for all intermediate years.

3.3 Step 2: Calculation & Validation

The MFA calculation uses an iterative solver:

```
mfa_system_results = run_mfa_calculation(mfa_system, ...)
```

The solver repeatedly passes through all processes until no flow values change between iterations (convergence). In each pass:

1. For each flow, the solver checks if the source process has calculated inflows
2. If yes, it applies the appropriate process logic (Splitter, Transformer, DSM, FOMP)
3. Transfer coefficients determine how inflow is distributed to outflows
4. Final balances are calculated: $\Delta S = \text{inflow} - \text{outflow}$ for each process

After calculation, the mass balance is displayed:

```
Mass Balance Check (material):
  Process 1 (Forestry):         Error = 0.00e+00
  Process 2 (Processing):       Error = 0.00e+00
  Process 3 (Use Phase):        Error = 2.13e-10
```

Errors below 10^{-6} are normal (floating-point precision). Larger errors indicate a configuration problem.

3.4 Step 3: Visualization

BioDYM provides interactive Plotly-based visualizations (see Chapter 6 for details):

- **Sankey diagram** — Overview of all flows, filterable by element and year
- **Process dynamics** — Inflow, outflow, and stock change per process over time
- **Flow dynamics** — Time series of individual flows
- **Stock analysis** — Bar charts and time series of accumulated stocks
- **Composition analysis** — Element composition of flows and stocks
- **DSM stock details** — Per-category stock evolution for DSM processes

3.5 Step 4: Export & Advanced Analysis

Results can be exported and further analyzed:

- **Excel export** — All flows, stocks, and parameters to `.xlsx`
- **KPI dashboard** — Key performance indicators (total input, output, stock, recycling rates)
- **Monte Carlo** — Uncertainty analysis (Chapter 9)

- **Scenarios** — Parameter comparison studies (Chapter 10)

Chapter 4

Excel Configuration

All model inputs are defined in a single Excel workbook (`.xlsm` or `.xlsx`). This chapter describes each sheet and its required columns.

4.1 Sheet Overview

Sheet Name	Purpose	Required
`Configuration`	Time range, elements, flags	Yes
`1_1_Definition_Flows`	Flow definitions and connections	Yes
`1_2_Data_Flows`	Flow time series data	Yes
`2_1_Definition_Processes`	Process definitions and logic	Yes
`2_2_static_TCs`	Static transfer coefficients	Optional
`2_3_dynamic_TCs`	Time-varying transfer coefficients	Optional
`2_4_Initial_Stock`	Initial stock values	Optional
`3_1_Definition_DSM`	DSM parameters	If DSM used
`3_2_Definition_FOMP`	FOMP parameters	If FOMP used
`4_1_Uncertainty_Parameters`	Monte Carlo distributions	If MC used
`5_1_Scenario_Definitions`	Scenario configurations	If scenarios used

Table 4.1: Excel workbook sheets.

4.2 Configuration Sheet

Defines global model settings.

Parameter	Example	Description
`Start_Year`	2020	First year of the model
`End_Year`	2050	Last year of the model
`Elements`	material,WC,DM,CC	Comma-separated element list
`Element_ID_X`	CC	Element name for hierarchy
`Parent_Element_ID_X`	DM	Parent element name

Table 4.2: Configuration sheet parameters.

4.3 Flow Definitions (Sheet 1_1)

Defines all flows in the system: their source, destination, and element composition.

4.3.1 Required Columns

Column	Example	Description
`Flow_ID`	`F_01_02`	Unique flow identifier
`Source_Process`	1	ID of the starting process
`Target_Process`	2	ID of the ending process
`E1_name`	WC[%]	First element name (composition fraction)
`E1_value`	60	Fraction of material that is this element (%)
`E2_name`	CC_DM[%]	Second element (CC as % of DM)
`E2_value`	45	Fraction value

Table 4.3: Flow definition columns.

4.3.2 Naming Convention

Flow IDs follow the pattern `F_<source>_<target>_<name>`:

- `F_01_02` — Flow from Process 1 to Process 2
- `F_05_00_CO2` — CO_2 flow from Process 5 to the environment

4.4 Flow Data (Sheet 1_2)

Contains time series values for input flows (flows from the system boundary).

Flow_ID	Year	E1_value	Unit
`F_00_01`	2020	1000	Mg
`F_00_01`	2030	1500	Mg
`F_00_01`	2050	2000	Mg

Table 4.4: Flow data example. Years between data points are interpolated linearly.

Tip

You only need to provide data for key years. BioDYM interpolates linearly between them. For example, defining values at 2020, 2030, and 2050 will automatically fill all intermediate years.

4.5 Process Definitions (Sheet 2_1)

Defines each process node in the system.

4.5.1 Process Logic Values

- **Input** — System boundary (Process 0); source of input flows

Column	Example	Description
`ID`	3	Unique process number
`Name`	Use_Phase	Human-readable name
`Process_Logic`	Splitter	How the process handles material
`Stock_Configuration`	Stock	Whether the process has a stock

Table 4.5: Process definition columns.

- **Splitter** — Distributes inflow to outflows by TCs; preserves composition
- **Transformer** — Like Splitter, but allows element-specific TCs (changes composition)
- **Pass-through** — Passes all material directly to a single outflow
- **DSM** — Dynamic stock model with lifetime distributions
- **FOMP** — First-order multi-pool organic matter decay

4.5.2 Stock Configuration Values

- **NoStock** — Process has no stock
- **Stock** — Process accumulates stock (zero initial stock)
- **Stock_with_InitialStock_Decay** — Stock with initial value, simple exponential decay
- **Stock_with_InitialStock_Cohort** — Stock with initial value, ODYM age-cohort method

4.6 Transfer Coefficients

TCs control how material leaving a process is distributed among its outflows.

4.6.1 Static TCs (Sheet 2_2)

Constant values that don't change over time:

TC_ID	Value	Unit
`TC_05_06`	0.7	1
`TC_05_07`	0.3	1

Table 4.6: Static TC example. Outflows from Process 5: 70% to Process 6, 30% to Process 7.

4.6.2 Dynamic TCs (Sheet 2_3)

Time-varying values, defined at key years and interpolated:

4.6.3 TC Naming Convention

- **Standard**: `TC_<source>_<target>` — e.g., `TC_05_06`
- **Element-specific**: `TC_E<idx>_<source>_<target>` — e.g., `TC_E2_05_06` for the WC-specific TC

TC_ID	Year	Value
`TC_05_06`	2020	0.6
`TC_05_06`	2040	0.8
`TC_05_07`	2020	0.4
`TC_05_07`	2040	0.2

Table 4.7: Dynamic TC example. Values interpolated between 2020 and 2040.

Element-specific TCs are only used with *Transformer* processes. Splitters use the same TC for all elements.

Warning

TCs for all outflows of a process must sum to 1.0 (100%) at every time step. BioDYM normalizes automatically if they don't, but will print a warning if the deviation exceeds 5%.

4.7 Initial Stock (Sheet 2_4)

Defines starting stock values for processes with stock configuration:

Process_ID	Parameter_Name	Parameter_Value	Unit
1	Initial_Stock_material	5000	Mg
1	Initial_Stock_WC[%]	15	%
1	Initial_Stock_DM[%]	85	%
1	Initial_Stock_CC[%]	45	%
1	Annual_Consumption_Rate	0.05	1/year

Table 4.8: Initial stock configuration example.

Chapter 5

Running the Calculation

5.1 The Iterative Solver

BioDYM uses an iterative convergence solver. In each pass, it processes all flows in the system:

1. Check if a flow's source process has calculated inflows available
2. If yes, apply the process logic (Splitter, Transformer, DSM, FOMP)
3. Compute outflow values using transfer coefficients
4. Repeat until no flow values change between passes

This approach handles circular flows (recycling loops) naturally — each pass propagates values further through the system until convergence.

```
mfa_system_results = run_mfa_calculation(
    mfa_system,
    dsm_params_config=dsm_params,       # Optional: DSM parameters
    fomp_params_config=fomp_params,      # Optional: FOMP parameters
    process_logic_map=process_logic_map,
    flow_tc_map=flow_tc_map,
    initial_stock_configs=initial_stock_configs,
)
```

5.2 Process Logic Details

5.2.1 Splitter

Distributes the total inflow to outflows using transfer coefficients. All elements maintain their input composition:

$$F_{\text{out},e} = F_{\text{in},e} \times \text{TC}_{\text{material}} \tag{5.1}$$

where e is the element index and TC is the material-level transfer coefficient.

5.2.2 Transformer

Like Splitter, but allows element-specific TCs. The composition of the outflow can differ from the inflow:

$$F_{\text{out},e} = F_{\text{in},e} \times \text{TC}_e \tag{5.2}$$

where TC_e can be different for each element.

5.2.3 Pass-through

Passes all inflow directly to a single outflow without modification:

$$F_{\text{out}} = F_{\text{in}} \tag{5.3}$$

5.3 Final Balances

After convergence, the solver calculates the stock change for every process:

$$\Delta S_p = \sum F_{\text{in},p} - \sum F_{\text{out},p} \tag{5.4}$$

For DSM processes, the stock S is calculated by the DSM engine (cohort tracking), and the solver preserves it. For all other processes, ΔS is the algebraic residual.

5.4 Mass Balance Verification

After calculation, a mass balance check is displayed for each process and element:

$$\text{Balance}_p = \sum F_{\text{in},p} - \sum F_{\text{out},p} - \Delta S_p \tag{5.5}$$

This should be zero (or very close to zero, within floating-point tolerance $\sim 10^{-10}$) for every process.

Warning

If you see mass balance errors larger than 10^{-3} Mg, check your transfer coefficients and flow definitions. Common causes:

- TCs not summing to 1.0 for a process
- Missing outflows for a process with inflows
- Incorrect element composition percentages

5.5 Element Hierarchy Recalculation

After TC application, hierarchical elements are recalculated to maintain consistency. For example, if CC is defined as a percentage of DM, the solver ensures:

$$\text{CC}_{\text{out}} = \text{DM}_{\text{out}} \times \frac{\text{CC}_{\text{in}}}{\text{DM}_{\text{in}}} \tag{5.6}$$

This prevents "transmutation" of elements — each flow maintains physically consistent composition.

Chapter 6

Visualization

BioDYM provides interactive Plotly-based visualizations in the Jupyter notebook. All plots follow a consistent publication style and can be exported as PNG (150 or 300 DPI), PDF, or SVG.

6.1 Sankey Diagram

The Sankey diagram provides a system-level overview of all material flows. Interactive controls:

- **Element selector** — Switch between material, WC, DM, CC
- **Year slider** — View flows for any year in the model range
- **Threshold** — Hide flows below a minimum value

Flow widths are proportional to mass. Node colors follow the process type color scheme.

6.2 Process Dynamics (3-Panel View)

For each process, a three-panel plot shows:

1. **Top panel**: Inflows over time
2. **Middle panel**: Outflows over time
3. **Bottom panel**: Stock change (ΔS) over time

Select the process and element via dropdown menus.

6.3 Flow Dynamics

Time series plots of individual flows. Useful for verifying that input data was loaded correctly and that calculated flows behave as expected.

6.4 Stock Analysis

- **Stock bar chart** — Stacked bar showing stock levels across all processes at a selected year
- **System stock composition** — Element breakdown of total system stock over time

6.5 DSM Stock Details

For processes with DSM logic, a specialized plot shows:

- Stock evolution stacked by category (e.g., construction wood types)
- Selectable by process and element
- Includes decaying initial stock contribution

See Chapter 7 for details on DSM configuration.

6.6 Composition Analysis

Interactive composition explorer showing the element fractions (WC, DM, CC) for each flow at each time step. Useful for verifying that compositions are physically consistent.

6.7 Exporting Plots

Each interactive plot has an export button or can be exported programmatically:

```
fig.write_image("my_plot.png", width=1200, height=800, scale=2)
```

Tip

For publication-quality figures, use `scale=3` (300 DPI at 1200px width). The publication style module automatically applies consistent fonts, colors, and layout.

6.8 Color Scheme

BioDYM uses a consistent color palette:

Purpose	Color	Hex Code
Material element	Blue	`#2E86AB`
WC element	Green	`#28A745`
DM element	Gray	`#6C757D`
CC element	Yellow	`#FFC107`
Splitter process	Blue	`#2E86AB`
Transformer process	Pink	`#A23B72`

Table 6.1: Standard color palette.

Chapter 7

Dynamic Stock Model (DSM)

7.1 Concept

The Dynamic Stock Model tracks material that enters a process and remains in stock for a defined lifetime before leaving as outflow. It uses *cohort-based tracking*: each year's inflow is a separate cohort that decays according to a lifetime distribution.

$$S(t) = \sum_{\tau=0}^{t} I(\tau) \cdot [1 - F_{\text{CDF}}(t - \tau)] \tag{7.1}$$

where $S(t)$ is the stock at time t, $I(\tau)$ is the inflow in year τ, and F_{CDF} is the cumulative distribution function of the lifetime.

7.2 Configuration (Sheet 3_1)

DSM parameters are defined in the `3_1_Definition_DSM` sheet using a key–value format:

Parameter	**Example**	**Description**
`Process_ID`	7	Process to apply DSM to
`DSM_Inflow_Split_Cat_1`	0.6	60% of inflow to Category 1
`DSM_Inflow_Split_Cat_2`	0.4	40% of inflow to Category 2
`DSM_Lifetime_Type_Cat_1`	Normal	Distribution type for Cat. 1
`DSM_Lifetime_Mean_Cat_1`	30	Mean lifetime (years)
`DSM_Lifetime_StdDev_Cat_1`	5	Standard deviation (years)
`DSM_Lifetime_Type_Cat_2`	Weibull	Distribution type for Cat. 2
`DSM_Lifetime_Mean_Cat_2`	50	Mean lifetime (years)
`DSM_Lifetime_Shape_Cat_2`	2.5	Weibull shape parameter
`DSM_Category_Name_Cat_1`	Short-lived	Display name
`DSM_Category_Name_Cat_2`	Long-lived	Display name

Table 7.1: DSM parameter configuration.

7.3 Categories

DSM supports multiple categories per process. Each category receives a fraction of the total inflow (`Inflow_Split`) and has its own lifetime distribution. This models heterogeneous products — for example, construction wood split into short-lived (interior) and long-lived (structural) applications.

> **Note**
>
> The inflow splits must sum to 1.0 across all categories.

7.4 Lifetime Distributions

Five distribution types are available:

7.4.1 Fixed

All items leave stock at exactly the mean lifetime. Parameters: `Mean`.

7.4.2 Normal

Symmetric bell curve around the mean. Parameters: `Mean`, `StdDev`.

> **Warning**
>
> If `StdDev` exceeds 80% of `Mean`, some lifetimes become negative. Use *FoldedNormal* instead.

7.4.3 FoldedNormal

Like Normal, but negative values are folded to positive. Eliminates the negative lifetime problem. Parameters: `Mean`, `StdDev`.

7.4.4 LogNormal

Right-skewed distribution where some items last much longer than average. Always positive. Parameters: `Mean`, `StdDev` (of the lognormal curve, not of ln).

7.4.5 Weibull

Flexible distribution controlled by the `Shape` parameter:

- Shape < 1: Decreasing failure rate (infant mortality)
- Shape $= 1$: Constant failure rate (exponential decay)
- Shape > 1: Increasing failure rate (wear-out failures)
- Shape ≈ 3.6: Approximates a normal distribution

Parameters: `Mean`, `Shape`.

See Appendix A for the complete mathematical reference.

7.5 DSM Outflow Distribution

Outflows from DSM processes are distributed to downstream flows using the standard TC system. This means DSM outflow splits can be:

- Static (constant fractions)

- Dynamic (time-varying, from `2_3_dynamic_TCs`)
- Subject to Monte Carlo sampling

7.6 Initial Stock

DSM processes can start with an existing stock that decays over time. Two methods are available:

- **Exponential decay**: Simple decay based on mean lifetime
- **ODYM age-cohort**: Rigorous method that reconstructs the age distribution of the initial stock

Configure via `Stock_Configuration` in the process definition sheet.

Chapter 8

First-Order Multi-Pool (FOMP)

8.1 Concept

The First-Order Mineralization Process model simulates organic matter decomposition using two pools with different decay rates:

- **Labile pool** — Easily degradable fraction, fast decay ($k_{\text{labile}} \approx 0.1$–$1.0$/year)
- **Recalcitrant pool** — Resistant fraction, slow decay ($k_{\text{recalcitrant}} \approx 0.01$–$0.05$/year)

Each pool follows first-order kinetics:

$$\frac{dS_i}{dt} = I_i(t) - k_i \cdot S_i(t) \tag{8.1}$$

where S_i is the stock in pool i, I_i is the inflow to pool i, and k_i is the decay rate constant.

The analytical solution for each time step is:

$$\text{Decay}_i(t) = S_i(t) \cdot \left(1 - e^{-k_i}\right) \tag{8.2}$$

8.2 Configuration (Sheet 3_2)

FOMP parameters are defined in the `3_2_Definition_FOMP` sheet:

Parameter	Example	Description
`Process_ID`	4	Process to apply FOMP to
`f_labile`	0.7	Fraction of inflow to labile pool
`k_labile`	0.5	Labile decay rate (1/year)
`k_recalcitrant`	0.025	Recalcitrant decay rate (1/year)
`outflow_id`	F_04_00_CO2	Carbon outflow (mineralization)
`outflow_id_2`	F_04_00_env	Environmental outflow (non-carbon)

Table 8.1: FOMP parameter configuration.

Note

The recalcitrant fraction is automatically calculated as $f_{\text{recalcitrant}} = 1 - f_{\text{labile}}$.

8.3 Element Handling

FOMP processes handle elements differently based on their nature:

Carbon outflow Contains the mineralized carbon mass. Elements: material = DM = CC = carbon mass.

Environmental outflow Contains the non-carbon dry matter and bypassed water. Elements calculated as:

- WC = water content of inflow (bypasses decay)
- DM = non-carbon dry matter decay
- CC = 0 (all carbon goes to carbon outflow)

8.4 Use Cases

FOMP is appropriate for modeling:

- Composting processes (labile fraction decays in weeks, recalcitrant in years)
- Soil organic matter turnover
- Anaerobic digestion (with adjusted rate constants)
- Landfill degradation (very slow recalcitrant decay)

Warning

FOMP uses a time-averaged CC/DM ratio as a constant for element calculations. This approximation can cause a small mass balance error (~2%) for the CC element. This is a known modeling limitation, not a bug.

Warning

FOMP treats the water content (WC) of the inflow as a passive bypass — water is not consumed by the decay process. This assumption is physically correct for **aerobic** systems (composting, soil organic matter turnover) where oxygen comes from outside the system boundary. For **anaerobic** landfill conditions, water is a reactant in methanogenesis and is incorporated into CH_4 and CO_2. Using FOMP for anaerobic processes introduces a methodological error in the WC element balance. Use the LFG module (Chapter 12) for landfill applications.

Chapter 9

Monte Carlo Simulation

9.1 Concept

Monte Carlo (MC) simulation quantifies how uncertainty in input parameters propagates through the model. In each iteration:

1. Sample random values for selected parameters from their defined distributions
2. Apply the sampled values to the MFA system
3. Run the solver
4. Record the results

After N iterations, the distribution of results shows the uncertainty range for all flows and stocks.

9.2 Configuration (Sheet 4_1)

Uncertainty parameters are defined in the `4_1_Uncertainty_Parameters` sheet:

Parameter_Name	**Distribution**	**Mean**	**StdDev**	**Min**	**Max**
`TC_05_06`	normal	0.65	0.05	0.4	0.9
`F_00_01`	uniform	–	–	900	1100
`TC_03_04`	triangular	–	–	0.3	0.5

Table 9.1: MC parameter configuration. The `Mode` column (not shown) is required for triangular distributions.

9.2.1 Available Distributions

normal Symmetric uncertainty. Parameters: `Mean`, `StdDev`. Optional: `Min`, `Max` for bounds.

uniform Equal probability within range. Parameters: `Min`, `Max`.

triangular Asymmetric with known most-likely value. Parameters: `Min`, `Max`, `Mode`.

lognormal Always positive, right-skewed. Parameters: `Mean`, `StdDev`. Optional: `Min`, `Max`.

9.3 Running MC Simulation

In the notebook:

```
mc_results = run_monte_carlo(
    mfa_system_configured=mfa_system,
    uncertainty_params=uncertainty_params,
    n_iterations=100,
    process_logic_map=process_logic_map,
    flow_tc_map=flow_tc_map,
    dsm_params_config=dsm_params,
    fomp_params_config=fomp_params,
)
```

Tip

Start with 100 iterations to verify the setup, then increase to 500–1000 for publication-quality results. Each iteration runs the full solver, so computation time scales linearly.

9.4 TC Normalization

When MC samples modify individual TCs, the outflow splits may no longer sum to 1.0. BioDYM uses *rejection sampling*: if sampled TCs deviate too far from 100%, the sample is rejected and redrawn. This preserves mass balance while respecting the defined uncertainty ranges.

9.5 Results

MC results include:

- **Summary statistics** — Mean, standard deviation, confidence intervals for all flows and stocks
- **Histograms** — Distribution of key output variables across iterations
- **Sensitivity analysis** — Tornado chart showing which parameters have the largest effect
- **Simulation paths** — Spaghetti plots showing all iterations overlaid
- **Mass balance check** — Per-process mass balance error across iterations (in scientific notation)

Chapter 10

Scenario Analysis

10.1 Concept

The scenario engine allows comparing different "what-if" situations against a baseline. Each scenario defines a set of parameter modifications that are applied before re-running the solver. All scenarios share the same system structure — only parameter values change.

10.2 Configuration (Sheet 5_1)

Scenarios are defined in the `5_1_Scenario_Definitions` sheet:

Scenario	Parameter	Type	Value	Start	End
Scenario_A	TC_05_06	multiply	1.2	2030	2050
Scenario_A	F_00_01	add	200	2025	2040
Scenario_B	TC_05_06	set	0.9	2030	2050

Table 10.1: Scenario definition example.

10.2.1 Modification Types

set Replace the parameter with a fixed value

multiply Multiply the current value by a factor

add Add a value to the current parameter

10.2.2 Time Range

Each modification specifies a `Start_Year` and `End_Year`. The modification is only applied within this range. Outside the range, the baseline value is used.

10.3 Supported Parameters

The scenario engine auto-detects the parameter type from the naming pattern:

- **Transfer coefficients** — Names starting with `TC_`
- **Flow data** — Names starting with `F_`
- **DSM parameters** — Names containing `DSM_`

- **FOMP parameters** — Names containing `FOMP_`

10.4 Running Scenarios

```
all_scenario_results = run_all_scenarios(
    mfa_system_configured=mfa_system,
    scenario_definitions=scenario_definitions,
    process_logic_map=process_logic_map,
    flow_tc_map=flow_tc_map,
    dsm_params_config=dsm_params,
    fomp_params_config=fomp_params,
)
```

Each scenario creates a deep copy of the baseline system, applies modifications, and re-runs the solver. Results are stored per scenario for comparison.

10.5 Comparison Plots

The scenario comparison visualization provides:

- **Flow dynamics** — Baseline vs. scenario flows over time
- **Stock dynamics** — Baseline vs. scenario stocks over time
- Multi-scenario selection (compare 2+ scenarios simultaneously)
- Element selector (view any tracked element)

Tip

Use scenarios to evaluate policy options (e.g., increased recycling rates), technology changes (e.g., new processing efficiency), or demand projections (e.g., reduced input flows).

Chapter 11

Results Export

11.1 KPI Dashboard

The KPI dashboard exports key performance indicators to `01_data/02_output/kpi_dashboard/system_kp`

- Total system input (all elements)
- Total system output (all elements)
- Net stock change
- Cumulative input/output totals
- Per-process stock levels

11.2 Results Excel Export

The full results are exported to `01_data/02_output/results_scientific_baseline.xlsx`:

```
export_results_to_excel(
    mfa_system_results,
    output_path="01_data/02_output/results_scientific_baseline.xlsx"
)
```

The export contains:

- **Flows sheet** — All flow values for all elements over time
- **Stocks sheet** — Stock levels (S) and stock changes (ΔS) per process
- **Parameters sheet** — All input parameters used in the calculation

11.3 Composition Export

Element composition data is exported to `01_data/02_output/composition_export/flow_composition.xl` containing the composition fractions (WC%, DM%, CC%) for every flow at every time step.

11.4 Plot Export

Individual plots can be exported via the export button on each interactive widget, or programmatically:

```
# PNG at 300 DPI
fig.write_image("plot.png", width=1200, height=800, scale=3)

# PDF (vector)
fig.write_image("plot.pdf", width=1200, height=800)

# SVG (vector, editable)
fig.write_image("plot.svg", width=1200, height=800)
```

11.5 Monte Carlo Results

MC results are displayed in the notebook via interactive plots. Summary statistics (mean, std, CI) for each variable are accessible programmatically from the `mc_results` dictionary.

11.6 Scenario Results

Scenario results are stored in the `all_scenario_results` dictionary, keyed by scenario name. Each entry contains a full solved MFA system that can be exported using the same `export_results_to_excel` function.

Chapter 12

Landfill Gas Model (LFG)

12.1 Theoretical Background

12.1.1 First-Order Decay — General Form

The Landfill Gas module uses the same first-order decay mathematics as the FOMP module (Chapter 8), extended to N user-defined waste fractions with individual decay constants and organic carbon fractions.

For each waste fraction j, the degradable organic carbon (DOC) stock evolves as:

$$\frac{dS_j}{dt} = I_j(t) - k_j \cdot S_j(t) \tag{12.1}$$

The analytical solution per time step is:

$$\text{Decay}_j(t) = S_j(t) \cdot \left(1 - e^{-k_j}\right) \tag{12.2}$$

where $S_j(t)$ is the organic carbon stock (Mg C) and k_j is the fraction-specific first-order decay constant (year^{-1}).

12.1.2 FOMP vs. LFG: Same Math, Different System

Aspect	**FOMP**	**LFG**
Pools	2 (labile + recalcitrant)	N user-defined fractions
Gas output	CO_2 via aerobic oxidation	CH_4 + biogenic CO_2
Water	Bypass (not consumed)	Reactant (H_2O consumed)
Carbon tracking	CC element (Mg C)	CC element (Mg C)
Residual	Recalcitrant pool	Organic C stock + ash

Table 12.1: Comparison of FOMP and LFG modules.

Warning

FOMP treats water content (WC) as a passive bypass — appropriate for aerobic processes where oxygen comes from outside the system boundary. In anaerobic landfill decomposition, H_2O is a reactant: water is consumed and hydrogen is incorporated into CH_4. For anaerobic systems, use the LFG module instead of FOMP. A known FOMP limitation is noted in Chapter 8.

12.2 UNFCCC AM-Tool-04 Methodology

12.2.1 Core FOD Equation

The LFG module implements the First-Order Decay (FOD) methodology from the UNFCCC Clean Development Mechanism AM-Tool-04 and the IPCC 2019 Refinement (Waste Sector, Volume 5, Chapter 3):

$$E_{\mathrm{CH_4}}(y) = \sum_j \sum_{x \le y} W_{j,x} \cdot \mathrm{DOC}_j \cdot \mathrm{DOC}_f \cdot F \cdot \mathrm{MCF} \cdot k_j \cdot e^{-k_j(y-x)} \cdot (1 - \mathrm{OX}) \tag{12.3}$$

In the BioDYM discrete time-step implementation this simplifies to:

$$I_j(t) = W(t) \cdot f_j \cdot \mathrm{DOC}_j \cdot \mathrm{DOC}_f \tag{12.4}$$

$$\mathrm{decay}_j(t) = S_j(t-1) \cdot \left(1 - e^{-k_j}\right) \tag{12.5}$$

$$S_j(t) = S_j(t-1) - \mathrm{decay}_j(t) + I_j(t) \tag{12.6}$$

$$C_{\mathrm{CH_4},j}(t) = \mathrm{decay}_j(t) \cdot F_{\mathrm{CH_4}} \cdot \mathrm{MCF} \cdot (1 - \mathrm{OX}) \tag{12.7}$$

$$C_{\mathrm{CO_2},j}(t) = \mathrm{decay}_j(t) \cdot (1 - F_{\mathrm{CH_4}}) \cdot \mathrm{MCF} \cdot (1 - \mathrm{OX}) \tag{12.8}$$

12.2.2 Waste Fraction Parameters

Each waste fraction is characterised by three parameters:

k_j First-order decay constant (year^{-1}). Depends on waste type and climate. IPCC 2019 temperate climate defaults are provided in Table 12.2.

DOC$_j$ Degradable Organic Carbon content (Mg C per Mg wet waste). Derived from waste composition analysis (see Section 12.4.1).

f_j Mass fraction of total waste input assigned to this fraction. Must sum to 1 across all fractions.

$f_{\mathbf{ash},j}$ Inert ash fraction (Mg ash per Mg waste). Stays in the landfill permanently; included in the stable stock.

12.2.3 Site-Level Correction Factors

12.3 Mass Balance and Element Handling

12.3.1 Anaerobic Decomposition — Water as Reactant

In anaerobic decomposition, water is consumed as a chemical reactant (not bypassed), because hydrogen atoms are incorporated into methane molecules:

Waste type	k **(year^{-1})**	DOC	**Source**
Food waste	0.185	0.15	IPCC 2019
Garden waste	0.100	0.20	IPCC 2019
Paper/cardboard	0.040	0.40	IPCC 2019
Wood/straw	0.020	0.43	IPCC 2019
Textile	0.040	0.24	IPCC 2019

Table 12.2: IPCC 2019 default parameters (temperate climate, $> 10\,°C$ annual mean temperature). For tropical climates, apply Arrhenius correction.

Parameter	**Name**	**Default**	**Description**
MCF	Methane correction factor	0.8	Accounts for site management type (managed anaerobic: 1.0; semi-aerobic: 0.5; open dump: 0.4)
DOC_f	Decomposable fraction	0.5	Fraction of DOC that actually decomposes under site conditions
$F_{\mathrm{CH_4}}$	CH_4 volume fraction	0.5	CH_4 share in raw landfill gas (typically 0.5–0.6)
OX	Oxidation factor	0.1	Fraction of CH_4 oxidised by cover soil before emission

Table 12.3: LFG site-level correction factors.

$$\text{Organic DM} + \mathrm{H_2O} \longrightarrow \mathrm{CH_4} + \mathrm{CO_2} \tag{12.9}$$

Approximate stoichiometry per Mg degraded organic dry matter:

$$1\,\mathrm{Mg\ org.\,DM} + 0.68\,\mathrm{Mg\,H_2O} \longrightarrow 0.59\,\mathrm{Mg\,CH_4} + 1.63\,\mathrm{Mg\,CO_2}$$

The total gas mass (2.22 Mg) exceeds the input organic dry matter (1 Mg) because water contributes mass to the products. This is why the WC element is consumed and exits as leachate, not bypassed.

12.3.2 Output Flows

CH_4 flow Tracked in Mg C (carbon mass in methane). BioDYM convention: `material = DM = CC = C_CH4` (identical to FOMP carbon outflow). To convert to actual CH_4 mass: multiply by 16/12.

CO_2 flow (biogenic) Tracked in Mg C. Same convention: `material = DM = CC = C_CO2`. Biogenic CO_2 is tracked but not counted as a greenhouse gas under standard accounting.

Leachate flow Water percolating through the landfill (Sickerwasser). `material = WC = WC_inflow`, `DM = 0`, `CC = 0`.

Stable stock Calculated by BioDYM mass balance: $S(t) = S(t-1) + \text{inflow} - \mathrm{CH_4} - \mathrm{CO_2} - \text{leachate}$. Contains residual undecayed organic carbon plus ash.

12.3.3 Carbon Stability and the Residual Stock

The stable stock is not pure inorganic matter — it contains:

- Undecayed organic carbon from all fractions $\left(\sum_j S_j(t)\right)$
- Inorganic ash $\left(\sum_j W(t) \cdot f_j \cdot f_{\text{ash},j}\right)$
- Non-degradable organic carbon (fraction $1 - \text{DOC}_f$ of input DOC)

This residual carbon represents long-term carbon storage, analogous to the recalcitrant pool in FOMP or the H/C_{org}-stable fraction in biochar (EBC stability classes). For carbon accounting, only the fraction that remains sequestered beyond the accounting boundary (typically 100 years) should be credited as stable.

12.4 Parameter Acquisition

12.4.1 DOC vs. TOC: Measurement Methods

The key distinction between Degradable Organic Carbon (DOC) and Total Organic Carbon (TOC):

TOC (Total Organic Carbon) All organic carbon in the waste, including both degradable and biologically stable fractions (lignin, charcoal, etc.). Measured by CHNS elemental analysis, loss-on-ignition (LOI, approximate), or wet oxidation (Walkley-Black).

DOC (Degradable Organic Carbon) The fraction of TOC that can be mineralised under landfill conditions. In IPCC methodology: `DOC = fraction of wet waste mass that is degradable organic C`. Typically $\text{DOC} \approx 0.5 \cdot \text{TOC}$ on a wet-weight basis.

The DOC_f parameter (fraction of DOC that actually decomposes) accounts for the fact that not all DOC is mineralised within the landfill lifetime, due to physical inaccessibility, pH inhibition, or microbial limitations.

12.4.2 Decay Constants: Literature, Lab, and Field Approaches

Three methods for deriving k_j values:

1. **IPCC defaults**: Use Table 12.2 as a starting point. Suitable for screening studies.
2. **Laboratory incubation**: Anaerobic batch incubation under controlled temperature and moisture. Fit the cumulative gas curve to $G(t) = G_\infty \cdot (1 - e^{-k\,t})$ to obtain k.
3. **Field back-calculation**: Use measured LFG collection data from an existing landfill and fit the FOD model to the observed gas curve.

12.4.3 Climate Correction (Arrhenius)

IPCC k values are given for temperate climate (10 °C mean annual temperature). Apply the Arrhenius correction for other climates:

$$k(T) = k_{\text{ref}} \cdot \exp\left[\frac{E_a}{R}\left(\frac{1}{T_{\text{ref}}} - \frac{1}{T}\right)\right] \tag{12.10}$$

where $E_a \approx 75\,\text{kJ/mol}$ is the activation energy for anaerobic decomposition, $R = 8.314\,\text{J/(mol K)}$, and temperatures are in Kelvin. IPCC 2019 provides pre-calculated climate zone multipliers as a simpler alternative.

12.5 Configuration (Sheet 3_3_Definition_LFG)

The sheet contains two row types, distinguished by the `LFG_Parameter_ID` field (format: `P{id}_{name}`):

Column	Example	Description
Waste fraction rows		
`LFG_Parameter_ID`	`P07_Food_waste`	Format: `P{process_id}_{fraction}`
`k_j`	0.185	Decay constant (year^{-1})
`DOC_j`	0.15	Degradable organic C (Mg C / Mg wet waste)
`f_input_j`	0.40	Mass fraction of waste input (must sum to 1)
`f_ash_j`	0.05	Inert ash fraction (Mg / Mg waste)
Site parameter rows (Value column)		
`P07_MCF`	0.8	Methane correction factor
`P07_DOCf`	0.5	Fraction of DOC that decomposes
`P07_F_CH4`	0.5	CH_4 volume fraction in LFG
`P07_OX`	0.1	Cover soil oxidation factor
`P07_outflow_ch4_id`	`F_07_00_CH4`	Flow ID for CH_4 output
`P07_outflow_co2_id`	`F_07_00_CO2`	Flow ID for biogenic CO_2 output
`P07_outflow_leachate_id`	`F_07_00_leachate`	Flow ID for leachate output

Table 12.4: LFG parameter configuration in sheet `3_3_Definition_LFG`.

Note

Set `Process_Logic = LFG` in the `2_1_Definition_Processes` sheet for each landfill process. The solver detects LFG processes by this label and calls `lfg_model.calculate_lfg()` automatically.

12.6 Known Limitations

- **Gas mass convention**: Flows are tracked in Mg C (carbon mass in the gas), not total gas mass. Multiply CH_4-C by 16/12 and CO_2-C by 44/12 to obtain mass in Mg.
- **Leachate dissolved organics**: Dissolved organic carbon (DOC) in leachate is currently set to zero (`DM = 0` in leachate flow). This is a conservative simplification; actual leachate DOC is typically small.
- **Initial stocks**: LFG processes start with zero carbon stock. For legacy landfills with significant existing waste, initial stocks should be estimated and applied via the `2_4_Initial_Stock` sheet (future enhancement).
- **Temperature dependence**: k_j values are assumed constant over the simulation period. Climate change effects on decay rates are not modelled automatically; use the Arrhenius correction for scenario sensitivity analysis.
- **Scenario analysis**: LFG processes are currently not included in the scenario engine. Scenario runs use only the baseline LFG configuration.

Appendix A

Distribution Functions Reference

This appendix provides the mathematical details for all distribution functions used in BioDYM.

A.1 DSM Lifetime Distributions

These distributions define how long material stays in stock before outflow.

A.1.1 Fixed

All items leave at exactly the mean lifetime.

$$f(t) = \delta(t - \mu) \qquad F(t) = \begin{cases} 0 & t < \mu \\ 1 & t \geq \mu \end{cases} \tag{A.1}$$

Parameters: `Mean` (μ).

A.1.2 Normal

$$f(t) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(t-\mu)^2}{2\sigma^2}\right) \tag{A.2}$$

Parameters: `Mean` (μ), `StdDev` (σ).

Use when lifetime variation is symmetric. Warning: produces negative lifetimes when $\sigma > 0.8\mu$.

A.1.3 FoldedNormal

$$f(t) = \frac{1}{\sigma\sqrt{2\pi}} \left[\exp\left(-\frac{(t-\mu)^2}{2\sigma^2}\right) + \exp\left(-\frac{(t+\mu)^2}{2\sigma^2}\right)\right], \quad t \geq 0 \tag{A.3}$$

Parameters: `Mean` (μ, before folding), `StdDev` (σ, before folding).

Eliminates negative lifetimes by folding the left tail onto the positive domain.

A.1.4 LogNormal

$$f(t) = \frac{1}{t \cdot \sigma_{\ln}\sqrt{2\pi}} \exp\left(-\frac{(\ln t - \mu_{\ln})^2}{2\sigma_{\ln}^2}\right), \quad t > 0 \tag{A.4}$$

where $\mu_{\ln}$ and $\sigma_{\ln}$ are derived from the user-provided Mean and StdDev:

$$\sigma_{\ln}^2 = \ln\left(1 + \frac{\sigma^2}{\mu^2}\right), \qquad \mu_{\ln} = \ln(\mu) - \frac{\sigma_{\ln}^2}{2} \tag{A.5}$$

Parameters: `Mean` (μ), `StdDev` (σ) of the lognormal curve (not of $\ln t$).

Right-skewed; always positive. Use for electronics, machinery, vehicles.

A.1.5 Weibull

$$f(t) = \frac{k}{\lambda}\left(\frac{t}{\lambda}\right)^{k-1} \exp\left(-\left(\frac{t}{\lambda}\right)^k\right), \quad t \geq 0 \tag{A.6}$$

where λ (scale) is derived from the user-provided Mean:

$$\lambda = \frac{\mu}{\Gamma(1 + 1/k)} \tag{A.7}$$

Parameters: `Mean` (μ), `Shape` (k).

Shape (k)	**Behavior**
$k < 1$	Decreasing failure rate (infant mortality)
$k = 1$	Constant failure rate (exponential)
$k = 2$	Linear increase (Rayleigh distribution)
$k \approx 3.6$	Approximates normal distribution
$k > 5$	Narrow, nearly fixed lifetime

Table A.1: Weibull shape parameter guide.

A.2 Monte Carlo Sampling Distributions

These distributions define uncertainty ranges for parameter sampling.

A.2.1 Normal (Bounded)

Samples from $\mathcal{N}(\mu, \sigma^2)$, optionally clipped to $[\text{Min}, \text{Max}]$.

Parameters: `Mean`, `StdDev`. Optional: `Min`, `Max`.

A.2.2 Uniform

Samples uniformly from $[\text{Min}, \text{Max}]$:

$$f(x) = \frac{1}{\text{Max} - \text{Min}}, \quad x \in [\text{Min}, \text{Max}] \tag{A.8}$$

Parameters: `Min`, `Max`.

A.2.3 Triangular

Samples from a triangle with peak at Mode:

$$f(x) = \begin{cases} \frac{2(x-a)}{(b-a)(c-a)} & a \leq x \leq c \\ \frac{2(b-x)}{(b-a)(b-c)} & c < x \leq b \end{cases} \tag{A.9}$$

where $a = \text{Min}$, $b = \text{Max}$, $c = \text{Mode}$.

Parameters: `Min`, `Max`, `Mode`.

A.2.4 LogNormal (Bounded)

Samples from a lognormal distribution, optionally clipped. Same parameterization as the DSM LogNormal.

Parameters: `Mean`, `StdDev`. Optional: `Min`, `Max`.

Appendix B

Excel Sheet Quick Reference

B.1 Sheet Summary

Sheet	Key Columns	Notes
`Configuration`	Start_Year, End_Year, Elements	Global settings. Elements as comma-separated list.
`1_1_Definition_Flows`	Flow_ID, Source_Process, Target_Process, E1_name, E1_value	One row per flow. E-columns define composition fractions.
`1_2_Data_Flows`	Flow_ID, Year, E1_value, Unit	Input flow time series. Gaps interpolated linearly.
`2_1_Definition_Processes`	ID, Name, Process_Logic, Stock_Configuration	One row per process. Logic: Input, Splitter, Transformer, Pass-through, DSM, FOMP.
`2_2_static_TCs`	TC_ID, Value, Unit	Constant TCs. Must sum to 1.0 per source process.
`2_3_dynamic_TCs`	TC_ID, Year, Value	Time-varying TCs. Interpolated and normalized automatically.
`2_4_Initial_Stock`	Process_ID, Parameter_Name, Parameter_Value	Key-value pairs: Initial_Stock_material, Initial_Stock_WC[%], Annual_Consumption_Rate.
`3_1_Definition_DSM`	Process_ID, Parameter, Value	Key-value pairs: Inflow_Split, Lifetime_Type/Mean/StdDev/Shape per category.
`3_2_Definition_FOMP`	Process_ID, Parameter, Value	Key-value pairs: f_labile, k_labile, k_recalcitrant, outflow_id.

Sheet	Key Columns	Notes
4_1_Uncertainty_Parameters	Parameter_Name, Distribution_Type, Mean, StdDev, Min, Max, Mode	MC distributions. Types: normal, uniform, triangular, lognormal.
5_1_Scenario_Definitions	Scenario_Name, Parameter_Name, Modification_Type, Value, Start_Year, End_Year	Modification types: set, multiply, add.

B.2 Naming Conventions

Type	Pattern	Example
Flow	`F_<src>_<dst>` or `F_<src>_<dst>_<name>`	`F_01_02`, `F_05_00_CO2`
TC (standard)	`TC_<src>_<dst>`	`TC_05_06`
TC (element)	`TC_E<idx>_<src>_<dst>`	`TC_E2_05_06`
Stock	`S_<process>`	`S_7`
Stock change	`dS_<process>`	`dS_7`

Table B.2: Naming patterns for system objects.

B.3 Process Logic Behavior

Logic	TC Type	Composition	Use Case
Input	None	Set by data	System boundary
Splitter	Material TC	Preserved	Distribution without transformation
Transformer	Element TCs	Changed	Chemical/physical conversion
Pass-through	None	Preserved	Direct connection
DSM	Material TC	Preserved	Stock with lifetime tracking
FOMP	Internal	Changed	Organic matter decay

Table B.3: Process logic comparison.

Appendix C

Troubleshooting

C.1 Installation Issues

"uv: command not found" Install uv: `curl -LsSf https://astral.sh/uv/install.sh | sh` (Linux/macOS) or see `https://docs.astral.sh/uv/` for Windows.

Tests fail after installation Run `uv sync` again to ensure all dependencies are installed. Check that Python 3.13+ is available: `python -version`.

Jupyter won't start Ensure Jupyter is installed: `uv run pip install jupyterlab`. Then retry `uv run jupyter lab`.

C.2 Data Loading Errors

"Sheet not found" The Excel workbook is missing a required sheet. Check that all sheet names match exactly (including underscores and case).

"Column X not found" A required column is missing or misspelled. Compare your sheet headers with the reference in Appendix B.

"No data for flow F_XX_YY" The flow is defined in sheet `1_1` but has no data in sheet `1_2`. Add at least one data point.

Decimal separator issues BioDYM expects the period (`.`) as decimal separator. If your Excel uses commas, either change the system locale or save as CSV with periods.

C.3 Calculation Errors

Solver does not converge Check for circular flows without a stock process. The iterative solver needs at least one stock (or boundary) to break circular dependencies.

Mass balance error $> 10^{-3}$

- TCs not summing to 1.0 — check all outflow TCs per process
- Missing outflow for a process that receives inflow
- Element composition fractions exceeding 100%

Negative stock values DSM lifetime parameters may be misconfigured. If using Normal distribution, check that `StdDev` $< 0.8\times$ `Mean`. Consider switching to FoldedNormal or Weibull.

DSM element stock drops to zero If non-material element stocks (WC, DM, CC) drop to zero when inflow stops, update to the latest BioDYM version. This was a known bug fixed with cumulative composition tracking.

C.4 Visualization Issues

Plots not showing in Jupyter Ensure `plotly` is installed: `uv run pip install plotly`. Use JupyterLab (not classic Jupyter Notebook) for best compatibility.

Export produces blank image Install `kaleido` for static image export: `uv run pip install kaleido`.

Small text on exported PNGs Increase the `scale` parameter: `fig.write_image("plot.png", scale=3)` for 300 DPI.

C.5 Monte Carlo Issues

MC always returns same results Check that the `Distribution_Type` column is spelled correctly (lowercase: `normal`, `uniform`, `triangular`, `lognormal`).

High rejection rate If many samples are rejected during TC normalization, your uncertainty ranges may be too wide. Narrow the `Min`/`Max` bounds or reduce `StdDev`.

C.6 Getting Help

- Check the test suite: `uv run python -m pytest 04_tests/ -v`
- Report issues: https://github.com/JScholz-tech/Biodym_JS/issues